

UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing
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LAB ACTIVITY

SECP 2523 DATABASE (WBL)

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LAB 1 - Creating a Database

CREATING THE TENNIS LOGOS DATABASE

Tennis Logos (TL) supplies customized tennis clothing and accessories for clubs and tournaments. The company purchases these items from suppliers at wholesale prices and adds the customer's logo. The final item price is determined by marking up the wholesale price and adding a fee that is based on the complexity of the logo.

Tennis Logos also does graphic design work for customers. Currently, the information about the items and the suppliers is stored in two Excel workbooks. Each item is assigned to a single supplier, but each supplier may be assigned many items.

You are to create a database that will store the item and supplier information. You already have determined that you need two tables, a Supplier table and an Item table, in which to store the information.

Instructions: *Perform the following tasks.*

1. *Use the Blank desktop database option to create a new database in which to store all objects related to the items for sale. Call the database – TL_Sport Database.*

```
mysql> CREATE DATABASE TL_Sport_Database;  
Query OK, 1 row affected (0.01 sec)  
mysql> USE TL_Sport_Database;  
Database changed
```

2. *Create the Supplier table using the structure shown in Figure 1. Use the name Supplier for the table.*

```
mysql> USE TL_Sport_Database;  
Database changed  
mysql> CREATE TABLE Supplier(  
-> supplierCode VARCHAR(2) PRIMARY KEY,  
-> supplierName VARCHAR(30),  
-> teleNo VARCHAR(12)  
-> );  
Query OK, 0 rows affected (0.02 sec)
```

3. Add the data shown in Figure 2 to the Supplier table.

```
mysql> INSERT INTO Supplier (supplierCode, supplierName, teleNo)
-> VALUES ('LM', 'Last Merchandisers', '803-555-7641')
-> ;
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Supplier (supplierCode, supplierName, teleNo)
-> VALUES ('PC', 'Pratt Clothing', '517-555-3853');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Supplier (supplierCode, supplierName, teleNo)
-> VALUES ('SD', 'Scryps Distributors', '610-555-8741');
Query OK, 1 row affected (0.00 sec)
```

4. Print the Supplier table.

```
mysql> SELECT * FROM Supplier;
+-----+-----+-----+
| supplierCode | supplierName      | teleNo      |
+-----+-----+-----+
| LM           | Last Merchandisers | 803-555-7641 |
| PC           | Pratt Clothing     | 517-555-3853 |
| SD           | Scryps Distributors | 610-555-8741 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

5. Create the Item table using the structure shown in Figure 3. Use the name Item for the table.

```
mysql> CREATE TABLE Item(
-> itemNumber VARCHAR(4) PRIMARY KEY,
-> description VARCHAR(20),
-> onHand INT(5),
-> wholesalePrice DECIMAL(10,2),
-> BaseCost DECIMAL(10,2),
-> supplierCode VARCHAR(2),
-> FOREIGN KEY (supplierCode) REFERENCES Supplier (supplierCode)
-> );
Query OK, 0 rows affected, 1 warning (0.04 sec)
```

6. Add the data shown in Figure 4 to the Item table.

```
mysql> INSERT INTO Item (itemNumber, description, onHand, wholesalePrice, BaseCost, supplierCode)
-> VALUES('3363', 'Baseball Cap', 110, 4.87, 6.10, 'LM'),('3673', 'Cotton Visor', 150, 4.59, 5.74,
'LM');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Item (itemNumber, description, onHand, wholesalePrice, BaseCost, supplierCode)
-> VALUES
-> ('4543', 'Crew Sweatshirt', 75, 7.29, 8.75, 'PC'),
-> ('4583', 'Crew T-Shirt', 135, 2.81, 3.38, 'PC'),
-> ('5923', 'Drink Holder', 75, 0.82, 1.07, 'SD'),
-> ('6185', 'Fleece Vest', 50, 28.80, 34.50, 'PC'),
-> ('6234', 'Foam Visor', 200, 0.79, 1.00, 'LM'),
-> ('6345', 'Golf Shirt', 115, 10.06, 12.35, 'PC'),
-> ('7123', 'Sports Bottle', 150, 1.04, 1.35, 'SD'),
-> ('7934', 'Tote Bag', 225, 1.26, 1.75, 'SD'),
-> ('8136', 'Travel Mug', 80, 2.47, 3.50, 'SD'),
-> ('8344', 'V-Neck Pullover', 76, 13.60, 15.75, 'LM'),
-> ('8590', 'V-Neck Vest', 65, 20.85, 25.02, 'LM'),
-> ('9458', 'Windbreaker', 54, 15.17, 18.20, 'PC'),
-> ('9620', 'Zip Hoodie', 90, 16.67, 20.04, 'PC');
Query OK, 13 rows affected (0.00 sec)
Records: 13 Duplicates: 0 Warnings: 0
```

7. *Print the Item table.*

```
mysql> SELECT * FROM Item;
```

itemNumber	description	onHand	wholesalePrice	BaseCost	supplierCode
3363	Baseball Cap	110	4.87	6.10	LM
3673	Cotton Visor	150	4.59	5.74	LM
4543	Crew Sweatshirt	75	7.29	8.75	PC
4583	Crew T-Shirt	135	2.81	3.38	PC
5923	Drink Holder	75	0.82	1.07	SD
6185	Fleece Vest	50	28.80	34.50	PC
6234	Foam Visor	200	0.79	1.00	LM
6345	Golf Shirt	115	10.06	12.35	PC
7123	Sports Bottle	150	1.04	1.35	SD
7934	Tote Bag	225	1.26	1.75	SD
8136	Travel Mug	80	2.47	3.50	SD
8344	V-Neck Pullover	76	13.60	15.75	LM
8590	V-Neck Vest	65	20.85	25.02	LM
9458	Windbreaker	54	15.17	18.20	PC
9620	Zip Hoodie	90	16.67	20.04	PC

```
15 rows in set (0.00 sec)
```

8. *Create a query for the Item table. Include the Item Number, Description, Wholesale Price, Base Cost, and Supplier Code. Save the query as Item Query.*

```
mysql> CREATE VIEW Item_Query AS
-> SELECT itemNumber, description, onHand, wholesalePrice, BaseCost, supplierCode
-> FROM Item;
Query OK, 0 rows affected (0.01 sec)
```

9. *Create a simple form for the Item table. Use the name Item Form for the form.*

```
mysql> CREATE VIEW Item_Form AS
-> SELECT itemNumber, description, onHand, wholesalePrice, BaseCost, supplierCode
-> FROM Item;
Query OK, 0 rows affected (0.01 sec)
```

10. *Drop the Supplier table.*

```
mysql> DROP TABLE Item;
Query OK, 0 rows affected (0.06 sec)

mysql> DROP TABLE Supplier;
Query OK, 0 rows affected (0.01 sec)

mysql> describe Supplier;
ERROR 1146 (42S02): Table 'tl_sport_database.supplier' doesn't exist
```

Structure of Supplier Table

<i>FIELD NAME</i>	<i>DATA TYPE</i>	<i>FIELD SIZE</i>	<i>PRIMARY KEY?</i>	<i>DESCRIPTION</i>
<i>Supplier Code</i>	<i>Short Text</i>	<i>2</i>	<i>Yes</i>	<i>Supplier Code (Primary Key)</i>
<i>Supplier Name</i>	<i>Short Text</i>	<i>30</i>		<i>Supplier Name</i>
<i>Telephone Number</i>	<i>Short Text</i>	<i>12</i>		<i>Supplier Telephone Number</i>

Figure 1**Data for Supplier Table**

<i>Supplier Code</i>	<i>Supplier Name</i>	<i>Telephone Number</i>
<i>LM</i>	<i>Last Merchandisers</i>	<i>803-555-7641</i>
<i>PC</i>	<i>Pratt Clothing</i>	<i>517-555-3853</i>
<i>SD</i>	<i>Scryps Distributors</i>	<i>610-555-8741</i>

Figure 2**Structure of Item Table**

<i>FIELD NAME</i>	<i>DATA TYPE</i>	<i>FIELD SIZE</i>	<i>PRIMARY KEY?</i>	<i>DESCRIPTION</i>
<i>Item Number</i>	<i>Short Text</i>	<i>4</i>	<i>Yes</i>	<i>Item Number (Primary Key)</i>
<i>Description</i>	<i>Short Text</i>	<i>20</i>		<i>Description of Item</i>
<i>On Hand</i>	<i>Number</i>			<i>On-Hand Item</i>
<i>Wholesale Price</i>	<i>Currency</i>			<i>Wholesale Price (Caption)</i>
<i>Base Cost</i>	<i>Currency</i>			<i>Base Cost for Item</i>
<i>Supplier Code</i>	<i>Short Text</i>	<i>2</i>		<i>Supplier Code</i>

Figure 3**Data for Item Table**

<i>Item Number</i>	<i>Description</i>	<i>On Hand</i>	<i>Wholesale Price</i>	<i>Base Cost</i>	<i>Supplier Code</i>
<i>3363</i>	<i>Baseball Cap</i>	<i>110</i>	<i>\$4.87</i>	<i>\$6.10</i>	<i>LM</i>
<i>3673</i>	<i>Cotton Visor</i>	<i>150</i>	<i>\$4.59</i>	<i>\$5.74</i>	<i>LM</i>
<i>4543</i>	<i>Crew Sweatshirt</i>	<i>75</i>	<i>\$7.29</i>	<i>\$8.75</i>	<i>PC</i>
<i>4583</i>	<i>Crew T-Shirt</i>	<i>135</i>	<i>\$2.81</i>	<i>\$3.38</i>	<i>PC</i>
<i>5923</i>	<i>Drink Holder</i>	<i>75</i>	<i>\$0.82</i>	<i>\$1.07</i>	<i>SD</i>
<i>6185</i>	<i>Fleece Vest</i>	<i>50</i>	<i>\$28.80</i>	<i>\$34.50</i>	<i>PC</i>
<i>6234</i>	<i>Foam Visor</i>	<i>200</i>	<i>\$0.79</i>	<i>\$1.00</i>	<i>LM</i>
<i>6345</i>	<i>Golf Shirt</i>	<i>115</i>	<i>\$10.06</i>	<i>\$12.35</i>	<i>PC</i>
<i>7123</i>	<i>Sports Bottle</i>	<i>150</i>	<i>\$1.04</i>	<i>\$1.35</i>	<i>SD</i>
<i>7934</i>	<i>Tote Bag</i>	<i>225</i>	<i>\$1.26</i>	<i>\$1.75</i>	<i>SD</i>
<i>8136</i>	<i>Travel Mug</i>	<i>80</i>	<i>\$2.47</i>	<i>\$3.50</i>	<i>SD</i>
<i>8344</i>	<i>V-Neck Pullover</i>	<i>76</i>	<i>\$13.60</i>	<i>\$15.75</i>	<i>LM</i>
<i>8590</i>	<i>V-Neck Vest</i>	<i>65</i>	<i>\$20.85</i>	<i>\$25.02</i>	<i>LM</i>
<i>9458</i>	<i>Windbreaker</i>	<i>54</i>	<i>\$15.17</i>	<i>\$18.20</i>	<i>PC</i>
<i>9620</i>	<i>Zip Hoodie</i>	<i>90</i>	<i>\$16.67</i>	<i>\$20.04</i>	<i>PC</i>

Figure 4